Photon — Cylindrical TauTimp wave equation with axial symmetry 19Mai2026 **photon.docx**

Author context. TauTimp framework, cosmological radius $R = 4.2355 \times 10^{22}$ m, speed of light $c = 2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$. Field treated as a scalar amplitude $H(r, \theta, z, t)$ with axial (cylindrical) symmetry, i.e. independent of θ.

1. Statement of the problem

Two coupled requirements are imposed:

(A) The cylindrical TauTimp wave equation for the field amplitude

$$\frac{R}{r}\,\Delta_{\mathrm{cyl}} f(r,z,t) \;=\; \frac{1}{c^2}\,\frac{\partial^2 f(r,z,t)}{\partial t^2}, \qquad f(R,0,t) = 0,$$

with cylindrical Laplacian (axisymmetric)

$$\Delta_{\mathrm{cyl}} f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial f}{\partial r}\right) + \frac{\partial^2 f}{\partial z^2} = f_{rr} + \frac{1}{r} f_r + f_{zz}.$$

(B) The Maxwell-type propagation equation for the squared-amplitude (probability) density $P = |\psi|^2$:

$$\frac{\partial^2 P(r,z,t)}{\partial t^2} \;-\; \frac{\partial^2 (P(r,z,t)\,c^2)}{\partial r^2} \;=\; 0, \qquad P(R,0,0) = 0,$$

with cylindrical symmetry, a Gaussian envelope in r, and zeros of $P(r,z,t_0)$ at t_0 fixed located on a periodic lattice $r = m\,r_b$, $m \in \mathbb{Z}_{\geq 0}$.

2. Equation (1): separation of variables

Write

$$f(r,z,t) \;=\; H_3(r)\,H_4(z)\,u(t).$$

Substituting into (1) and dividing by $H_3 H_4 u$:

$$\frac{R}{r}\left[\frac{H_3''}{H_3} + \frac{1}{r}\frac{H_3'}{H_3} + \frac{H_4''}{H_4}\right] \;=\; \frac{1}{c^2}\frac{u''}{u}.$$

2.1 Time piece

Right-hand side depends only on t. Set

$$\frac{u''}{u} = -\omega^2 \quad \Rightarrow \quad u(t) = \cos(\omega t + \varphi).$$

2.2 Axial piece

The z-direction is unbounded; choose oscillatory modes

$$\frac{H_4''}{H_4} = -k_z^2 \quad \Rightarrow \quad H_4(z) = A_z \cos(k_z z) + B_z \sin(k_z z).$$

2.3 Radial piece

Substituting (5)–(6) into (4):

$$\frac{R}{r}\left[\frac{H_3''}{H_3} + \frac{1}{r}\frac{H_3'}{H_3} - k_z^2\right] = -\frac{\omega^2}{c^2}.$$

Rearranging,

$$\boxed{H_3''(r) + \frac{1}{r}\,H_3'(r) + \left[\frac{\omega^2\, r}{R\,c^2} - k_z^2\right] H_3(r) = 0.}$$

This is the variable-coefficient radial equation (linear-in-r coefficient — the signature of the R/r prefactor).

3. Reduction to Airy form

Eliminate the first derivative via $H_3(r) = r^{-1/2}\,g(r)$. Substituting,

$$g''(r) + \left[\frac{\omega^2\, r}{R\,c^2} - k_z^2 + \frac{1}{4r^2}\right] g(r) = 0.$$

The $1/(4r^2)$ centrifugal-like term is subdominant for $r \gtrsim \sqrt{R\,c^2/\omega^2}$; we keep it implicit and analyze the leading two terms (a WKB-exact rearrangement; the $1/(4r^2)$ piece only contributes near $r = 0$).

Introduce the dimensionless coordinate

$$y \;=\; \kappa\left(r - \frac{k_z^2\, R\, c^2}{\omega^2}\right), \qquad \kappa \;=\; \left(\frac{\omega^2}{R\,c^2}\right)^{1/3}.$$

Then $\omega^2 r/(Rc^2) - k_z^2 = \kappa^2\, y$ and $g_{rr} = \kappa^2\, g_{yy}$. Equation (8) becomes (dropping $1/(4r^2)$):

$$g_{yy}(y) + y\,g(y) = 0 \quad \Leftrightarrow \quad g_{yy} = (-y)\,g.$$

This is the Airy equation in the argument $-y$, with general solution

$$g(y) = C_1\,\mathrm{Ai}(-y) + C_2\,\mathrm{Bi}(-y).$$

Regularity at large r (where $-y \to -\infty$, Bi diverges) forces $C_2 = 0$. Therefore

$$\boxed{H_3(r) \;=\; \frac{C_1}{\sqrt{r}}\,\mathrm{Ai}(-y(r)).}$$

4. Eigenvalue quantization from $f(R,0,t)=0$

The condition $f(R,0,t)=0\ \forall t$ with $u(t)\neq 0$ in general requires either $H_3(R)=0$ or $H_4(0)=0$. The non-trivial radial spectrum comes from

$$H_3(R)=0 \ \Leftrightarrow\ \mathrm{Ai}(-y(R))=0.$$

Let $\{a_n\}_{n\geq 1}$ be the (negative) zeros of Ai: $\mathrm{Ai}(a_n)=0$ with $|a_1|\approx 2.33810741$, $|a_2|\approx 4.08794944$, $|a_3|\approx 5.52055983$. Then (13) reads

$$-y(R)=a_n \ \Leftrightarrow\ y(R)=|a_n|,$$

which gives the dispersion relation

$$\boxed{\left(\frac{\omega_n^2}{R\,c^2}\right)^{1/3} R\left(1-\frac{k_z^2c^2}{\omega_n^2}\right) = |a_n|.}$$

For the planar mode $k_z=0$:

$$\left(\frac{\omega_n R}{c}\right)^{2/3}=|a_n| \quad\Rightarrow\quad \omega_n=\frac{c}{R}\,|a_n|^{3/2}.$$

Numerical values (using $c/R=7.0781\times 10^{-15}$ rad s^{-1}):

n	$\lvert a_n\rvert$	$\lvert a_n\rvert^{3/2}$	ω_n [rad s^{-1}]	period $T_n=2\pi/\omega_n$ [s]
1	2.33810741	3.57517	2.5305×10^{-14}	2.483×10^{14}
2	4.08794944	8.26399	5.8502×10^{-14}	1.074×10^{14}
3	5.52055983	12.96997	9.1810×10^{-14}	6.844×10^{13}
4	6.78670809	17.68252	1.2514×10^{-13}	5.021×10^{13}
5	7.94413359	22.39529	1.5848×10^{-13}	3.965×10^{13}

4.1 Full solution of (1)

$$\boxed{f_n(r,z,t)=\frac{C_n}{\sqrt{r}}\,\mathrm{Ai}(-y_n(r))\,[A_z\cos(k_z z)+B_z\sin(k_z z)]\cos(\omega_n t+\varphi),}$$

with $y_n(r)=\kappa_n(r-k_z^2Rc^2/\omega_n^2)$ and $\kappa_n=(\omega_n^2/(Rc^2))^{1/3}$.

4.2 Reconstructing $H(r,\theta,z,t)$

The full cylindrical field with axial (no-θ) symmetry is obtained by trivial extension; since (1) was solved with $\partial_\theta H=0$ and the cylindrical Laplacian above is the $m=0$ azimuthal sector, the general axisymmetric solution is

$$H(r,\theta,z,t)\ =\ \sum_{n,k_z} C_{n,k_z}\,\frac{\mathrm{Ai}\big(-y_{n,k_z}(r)\big)}{\sqrt{r}}\,\cos(k_z z+\phi_{k_z})\cos(\omega_n t+\varphi_n),$$

i.e. independent of θ as required.

5. Equation (2): Maxwell-type propagation and the Gaussian/periodic structure of P

Equation (2) is

$$\partial_{tt}P-c^2\,\partial_{rr}P=0,$$

(with c constant the c^2 pulls out of the spatial derivative). Any function of the single characteristic $\xi=r-ct$ (right-moving) or $\eta=r+ct$ (left-moving) solves (19). Take a right-moving packet with periodic carrier:

$$\psi(r,z,t)=N(z)\exp\left[-\frac{(r-ct)^2}{2\sigma_r^2}\right]\cos(k_r(r-ct)).$$

Since ψ depends on r,t only through ξ, ψ itself satisfies the (r,t) d'Alembertian: $\partial_{tt}\psi=c^2\,\partial_{rr}\psi$. Consequently

$$P(r,z,t)=|\psi|^2=N(z)^2\exp\left[-\frac{(r-ct)^2}{\sigma_r^2}\right]\cos^2(k_r(r-ct))$$

is also a pure function of ξ, hence

$$\boxed{\;\partial_{tt}P - c^{2}\,\partial_{rr}P \;=\; 0\;}\quad\text{(verified symbolically; SymPy returns 0).} \tag{22}$$

5.1 Periodic zeros of P at $r=m\,r_b$ for t_0 fixed

At any fixed t_0, zeros of $P(r,z,t_0)$ require $\cos(k_r(r-ct_0))=0$, i.e.

$$k_r(r-ct_0)=(m+1/2)\pi, \qquad m\in\mathbb{Z}_{\geq 0}.$$

Hence consecutive zeros are separated by

$$\Delta r=\frac{\pi}{k_r}\ \equiv\ r_b, \qquad\Rightarrow\qquad k_r=\frac{\pi}{r_b}.$$

The zero locations are $r_m=ct_0+(m+1/2)r_b$ — a periodic lattice with lattice spacing r_b, as required.

5.2 Boundary condition $P(R,0,0)=0$

Pick $\varphi_z=0$ for the axial profile and absorb it into $N(z)$. At $r=R$, $z=0$, $t=0$:

$$P(R,0,0)=N(0)^2\exp(-R^2/\sigma_r^2)\cos^2(k_rR)=0.$$

The Gaussian factor is positive; $N(0)\neq 0$. Therefore

$$\cos(k_rR)=0\ \Leftrightarrow\ k_rR=(n+1/2)\pi, \quad n\in\mathbb{Z}_{\geq 0}.$$

Combining (23)–(24): $\boxed{r_b = \frac{R}{n+1/2}, \qquad n = 0,1,2,\ldots}$

i.e. the lattice spacing r_b is itself quantized in units of R.

5.3 Matching to the lowest Airy eigenmode of (1)

If one identifies the carrier wavenumber of (20) with the natural radial wavenumber of the lowest TauTimp eigenmode, $k_r = \omega_1/c = |a_1|^{3/2}/R$:

$$r_b^{(\mathrm{nat})} = \frac{\pi}{k_r} = \frac{\pi R}{|a_1|^{3/2}} = \frac{\pi}{3.5752} R \approx 0.8787\, R \approx 3.7219 \times 10^{22}\ \mathrm{m}.$$

This value lies between the $n = 0$ ($r_b = 2R$) and $n = 1$ ($r_b = 2R/3$) admissible boundary-condition values: it is not exactly a BC-compatible value, indicating the variable-coefficient eigenproblem and the constant-c Maxwell limit are physically related but not identical. Strict satisfaction of $P(R, 0,0) = 0$ requires r_b from (25).

6. Summary table

Quantity	Symbol / form	Value or formula
TauTimp radius	R	4.2355×10^{22} m
Time scale	R/c	1.4128×10^{14} s
Radial separation eq.	(7)	$H_3'' + H_3'/r + (\omega^2 r/(Rc^2) - k_z^2)H_3 = 0$
Reduced equation	(10)	Airy $g_{yy} + yg = 0$
Radial eigenfunction	(12)	$H_{3,n}(r) = C_n r^{-1/2}\mathrm{Ai}(-y_n(r))$
Eigen-frequency ($k_z = 0$)	(16)	$\omega_n = (c/R)\,\lvert a_n\rvert^{3/2}$
Lowest mode	ω_1	2.5305×10^{-14} rad s^{-1}
Maxwell-eq. solution	(21)	$P = N^2 e^{-(r-ct)^2/\sigma_r^2}\cos^2(k_r(r - ct))$
Zero-lattice spacing	r_b	π/k_r
BC quantization	(25)	$r_b = R/(n + 1/2),\ n = 0,1,2,\ldots$
Natural match to mode-1	(26)	$r_b^{\mathrm{nat}} = \pi R/\lvert a_1\rvert^{3/2} \approx 0.8787\, R$

7. Physical interpretation

The variable coefficient R/r in (1) creates an inhomogeneous radial wave operator whose eigenfunctions are Airy functions — exactly the signature already encountered in the spherical TauTimp framework. In the cylindrical geometry, the $r^{-1/2}$ envelope (from the cylindrical metric factor) replaces the $1/r$ envelope seen in the spherical case, but the Airy-function quantization survives intact. The discrete frequencies $\omega_n = (c/R)|a_n|^{3/2}$ scale identically with $|a_n|^{3/2}$, confirming this is a robust feature of the R/r operator across geometries.

The companion Maxwell-type equation (2) describes the same photon as a localized wave packet on the same cosmological background. The Gaussian envelope sets the coherence width σ_r, and the cosine carrier sets the periodic node spacing r_b. Boundary condition $P(R, 0,0) = 0$ quantizes r_b in integer-plus-half submultiples of R (eqn. 25). Identifying the carrier k_r with the natural radial wavenumber ω_1/c of the lowest Airy eigenmode gives the non-quantized "free-space" value $r_b \approx 0.8787\, R$, which is the limit obtained when the strict boundary condition is relaxed. This is the cylindrical-coordinate counterpart of the spherical TauTimp photon result and is fully consistent with the doctoral framework's anchors (R, c, Airy quantization, $r^{-1/2}$ or r^{-1} amplitude envelope according to geometry).